MSH-Net: Modality-Shared Hallucination With Joint Adaptation Distillation for Remote Sensing Image Classification Using Missing Modalities

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***Abstract*— Learning-based multimodal data has attracted increasing interest in the remote sensing community owing to its robust performance. Although it is preferable to collect multiple modalities for training, not all of them are available in practical scenarios due to the restriction of imaging conditions. Therefore, how to assist the model inference with missing modalities is significant for multimodal remote sensing image processing. In this work, we propose a general framework called modality-shared hallucination network (MSH-Net) to address this issue by reconstructing complete modality-shared features from incomplete inference modalities. Compared to conventional privilege modality hallucination methods, MSH-Net does not only help preserve the cross-modal interactions for model inference but also scales well with the increasing number of missing modalities. We further develop a novel joint adaptation distillation (JAD) method that guides the hallucination model to learn the modality-shared knowledge from the multimodal model by matching the joint probability distributions between representation and groundtruth. This overcomes the representation heterogeneity caused by the discrepancy between inputs and structures of multimodal and hallucination model while preserving the decision boundaries refined by multimodal cues. Finally, extensive experiments conducted on four common modality combinations demonstrate that the proposed MSH-Net can effectively address the problem of missing modalities and achieve state-of-the-art performance. Code is available at: https://github.com/shicaiwei123/MSHNet**

***Index Terms*— Classification, deep learning, distillation knowledge, missing modalities, multimodal, remote sensing.**

I. Introduction

REMOTE sensing image classification plays an essential role in Earth observations, such as urban planning [1], land use analysis [2], environmental monitoring [3], etc. Recently, classification models that leverage multiple modalities information show great power in practice and gather growing attention from the research community [4], [5], [6], [7], [8], [9], [10], with the development of deep learning techniques. Although the multimodal methods show better performance over the unimodal ones, they may suffer from missing modality in real-world deployment scenarios due to the following two reasons: 1) sensor failures that lead to modality information loss completely [11], [12] and 2) the low sampling frequency of some sensors, such as satellites, raises the difficulty to obtain remote sensing images from the same areas in time [13]. Therefore, it is of great significance to deploy the multimodal remote sensing model in real-world scenarios where partial modalities are missing.

Generally, the information available only at the training stage is defined as privileged information [14]. Specifically, the data modality available only at the training stage is called privileged modality and the modality available at both training and inference stages is called regular modality. To date, enormous efforts have been made to deal with the multimodal classification with the privileged modality [13], [15], [16], [17], [18], and most methods can be classified into two types: auxiliary-based methods and generative-based methods. The auxiliary-based methods consider the privileged modality data as auxiliary input and leverage it to assist the optimization of decision boundaries via multitask learning, i.e., introducing an additional constraint term for privileged modality data [15]. On the other hand, the generative-based methods propose to restore the information of the privileged modality from the regular modality to assist the inference with missing modalities, since they share the same semantic information [13], [17], [18], [19], [20]. Because the auxiliary task does not necessarily satisfy a no-harm guarantee and may limit the performance of the primary task at hand [21], previous methods mainly explore the generative-based methods to aid the remote sensing image classification with missing modalities [13], [18], [20], [22].

According to the type of reconstructed targets, the generative-based methods can be further divided into two types: imputation-based methods and hallucination-based methods. The imputation-based methods reconstruct the privileged samples by imputation algorithms and then combine them with the regular ones as the input of multimodal models [16], [17], [19]. However, the sample reconstruction is unstable and may introduce harmful noise due to its

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complexity [23], [24], [25]. Therefore, the hallucination-based methods are proposed to handle the reconstruction issue from sample space to feature space [13], [18], [20]. They first train a hallucination model to reconstruct the privileged representation using distillation algorithms and then fuse the hallucination model and the regular model trained with regular modality data to make the final decision. Although these methods achieve good performance, they still suffer from two limitations. On the one hand, they fuse the information of privileged and regular modalities at the decision level. This includes only the modality-specific information and ignores the intermediate modality-shared properties that have been demonstrated as a crucial factor in multimodal remote sensing classification [5], [6], [7]. On the other hand, the number of hallucination branches of such methods increases with the number of missing modalities. This is inflexible and cumbersome, especially for scenarios with multiple missing modalities.

Meanwhile, knowledge distillation, as a knowledge transfer technique, has been widely used to train hallucination models. Previous hallucination-based methods mainly focus on the feature-based distillation (FD) methods that match intermediate representations directly [13], [18], [20], [26]. This would potentially lead to overfitting and limit the performance on unseen samples since it is not feasible for the hallucination model to generate the same feature map as the multimodal model due to their different inputs [27], [28]. As a possible solution, we can relax the constraints by replacing the FD methods with the relation-based distillation methods to train the hallucination model. This has been demonstrated to solve the representation heterogeneity problem effectively [29], [30], [31]. However, existing relation-based methods consider only the relationship between samples, which is not enough to guide the hallucination model to inherit the complete modality-shared information and may lead to suboptimal performance.

To address the above issues, we propose a general modality-shared hallucination network (MSH-Net) for remote sensing image classification with missing modalities. As shown in Fig. 1, the MSH-Net first trains the hallucination model to learn the modality-shared knowledge from the pretrained multimodal model directly and then fuses it with the regular model that extracts the modality-specific properties to make the final decision. Compared to conventional hallucination methods that learn from the model trained with privileged modality only, MSH-Net helps acquire the cross-modal interaction from the multimodal model. Moreover, since MSH-Net only aims to restore the complete multimodal knowledge instead of the missing information, it can scale with the increasing number of missing modalities. Second, we propose the joint adaptation distillation (JAD) method to transfer the modality-shared knowledge from the pretrained multimodal model to the hallucination model by matching their joint probability distribution of the sample representation and final prediction output. Beyond simply capturing the inter-representation relationship of the multimodal model, this further introduces the conditional distributions between the sample representation and final prediction output as an extra constraint to transfer the refined classification boundaries of the multimodal model, so that the hallucination model can inherit comprehensive multimodal knowledge and achieve improved performance. The major contributions of this work are summarized as follows.

1) We propose a novel MSH-Net to address the multimodal remote sensing image classification challenge with missing modality data by recovering the modality-shared features for model inference. This helps preserve the cross-modal interaction and improve the scalability for multiple missing modalities.
2) We propose a JAD method to transfer the multimodal knowledge to the hallucination model by matching the marginal and conditional distributions simultaneously. This overcomes the representation heterogeneity while acquiring comprehensive modality-shared information.
3) Experiments and ablation studies on four common modality combinations demonstrate the effectiveness and generalization of the proposed framework to infer with missing modalities.

II. Related Work

A. Generative-Based Methods

The generative-based methods can be divided into two types: imputation-based methods and hallucination-based methods. The imputation-based methods reconstruct the pixel information of the privileged modality from the available modality image for the pretrained multimodal classification model to handle the missing modality. For example, Zhu et al. [32] utilize the CycleGAN to generate the MRI image from the CT image to aid mediastinal lung tumor segmentation [19]. Due to the complexity and instability of sample reconstruction [24], [25], hallucination-based methods are proposed to reconstruct the representation information of the privileged modality. Kampffmeyer et al. [20] hallucinate the representation of the digital surface model (DSM) images from available RGB images to improve the optical land cover classification performance. Kumar et al. [18] hallucinate the representation of missing spectral data for multispectral (MS) land cover classification. Li et al. [13] introduce the dynamic-hierarchical attention distillation (DHAD) to hallucinate the synthetic aperture radar (SAR) image feature from RGB images to aid the land cover classification. Liu et al. [33] present a teacher assistant network to transfer the knowledge learned from RGB images to assist MS scene classification. Ni et al. [34] propose distance and angle-wised distillation to transfer RGB knowledge to wearable devices for human activity recognition. Hong et al. [35] introduce cross-modality knowledge distillation to transfer the light detection and ranging (LiDAR) information to the RGB model for autonomous driving. In summary, these works aim to train a hallucination model to reconstruct the representation of the privileged model by matching their intermediate feature maps directly. This may lead to overfitting and limit the performance of the hallucination model since the inputs of the hallucination and

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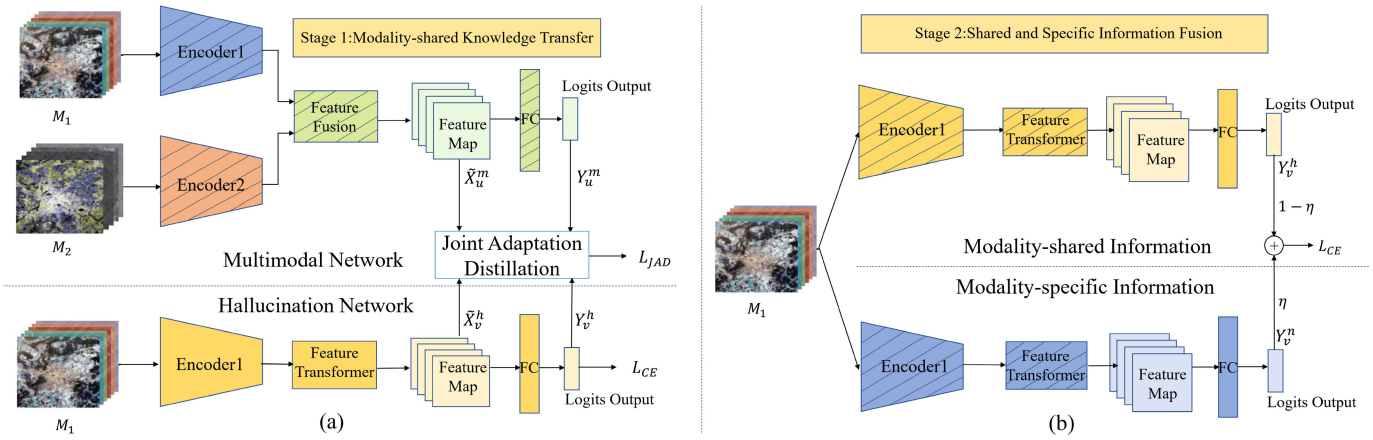


Fig. 1. Overall architecture of our MSH-Net, where M_1 and M_2 are two different inputs of the multimodal networks, and M_2 is the missing modality during inference. The pipeline of the MSH-Net consists of two stages: 1) the modality-shared knowledge transfer that guides the hallucination network to learn the shared knowledge among $\{M_1, M_2\}$ from the pretrained multimodal network via the JAD method and 2) the shared and specific information fusion that fuses the shared knowledge among the $\{M_1, M_2\}$ extracted in the hallucination network and the specific knowledge of the available modality M_1 extracted in the regular network to make the final decision. Specifically, the module with the slash lines means its parameters are frozen, i.e., the multimodal network in stage 1 and the feature encoders and transformer blocks in stage 2. The color means different information flows, blue—modality 1, orange—modality 2, green—fused information, yellow—hallucination information. Feature transformer and feature fusion modules share the same architecture except for the number of input channels. "FC" means fully-connected layer. The training procedure is described in Section III-F.

privileged networks are different and it is impossible to force them to generate the sample feature maps.

B. Knowledge Distillation

Knowledge distillation introduced by Hinton et al. [36] aims to transfer knowledge from a strong teacher to a weaker student network to facilitate supervised learning. Generally, it can be divided into three types: response-based distillation, representation-based distillation, and relation-based distillation. Response-based distillation transfers the neural response of the last output layer of the teacher to the student network by matching their softened logits output [36]. Although the representation-based distillation is straightforward and easy to understand, it fails to transfer the powerful representations learned by the teacher [37]. Therefore the representation-based distillation is proposed to model the knowledge of intermediate feature maps and transfer it to the student network by minimizing the discrepancy between the value [38], attention map [39], [40], and attention project [41] of their feature maps. Furthermore, relation-based distillation is proposed to model the relationships between different layers or data samples and transfer it by minimizing the discrepancy between the similarity map [29], distribution [42], interdata relations [43] of feature pairs. Although numerous relation-based knowledge distillation methods have been proposed, they consider only the representation relationship. This may be insufficient to guide the student model to learn complete information from the teacher models.

III. Methods

A. Overview

In this section, we will introduce the MSH-Net that guides the hallucination network to learn the modality-shared knowledge from the multimodal network directly. As shown in Fig. 1, the training pipeline of the MSH-Net consists of two stages: modality-shared knowledge transfer as well as shared and specific information fusion. In particular, to cope with the representation heterogeneity caused by the structure and input discrepancies, we propose the JAD method to transfer the modality-shared knowledge from the multimodal network to the hallucination network by matching their marginal and conditional distributions. This helps relax the optimization target, meanwhile guiding the hallucination model to learn full multimodal knowledge.

In summary, we will first introduce the notations used in our methods. Then we present the JAD strategy, modality-shared knowledge transfer, and shared and specific information fusion. Finally, we discuss the network training and testing details.

B. Notation

Let Θ denote the training dataset consisting of modalities $\{M_1, \ldots, M_{\mathcal{U}}\}$. Ψ denotes the dataset including only modalities $\{M_1, \ldots, M_{\mathcal{V}}\}$ that are available in the real-world application scenario. Here, dataset Ψ is generated from the dataset Θ by removing the missing modalities $\{M_{\mathcal{V}+1}, \ldots, M_{\mathcal{U}}\}$ for each sample. $X = (X_u, X_v)$ denotes a paired mini-batch sample with label Y, where X_u and X_v are chosen from the sample set Θ and Ψ, respectively. $\widetilde{X}_u^m = \varphi_m(X_u) \in R^{b\times c_m\times w_m\times t_m}$ and $\widetilde{X}_v^h = \varphi_h(X_v) \in R^{b\times c_h\times w_h\times t_h}$ denote the feature maps in the penultimate layer of the multimodal network and hallucination network, respectively. Here, b is the number of batch size, c is the number of feature channels, w and t are spatial dimensions. $Y_u^m = f_m(\widetilde{X}_u^m) \in R^{b\times l}$ and $Y_v^h = f_h(\widetilde{X}_v^h) \in R^{b\times l}$ denote logits output of the multimodal network and hallucination network, respectively. Here, l denotes the number of land cover categories. Finally, $\widetilde{X}_v^n = \varphi_n(X_v) \in R^{b\times c_n\times w_n\times t_n}$ and $Y_v^n = f_n(\widetilde{X}_v^n) \in R^{b\times l}$ denote the feature maps in the penultimate

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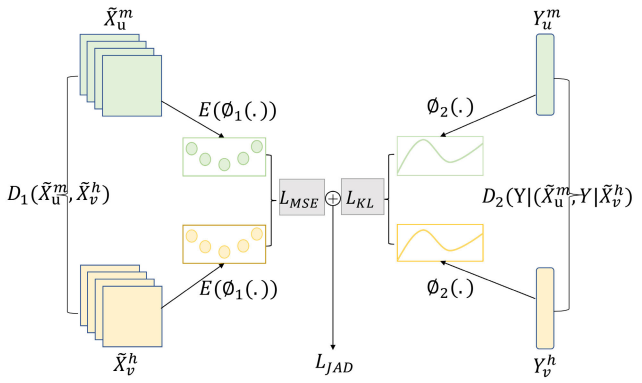


Fig. 2. Overall architecture of the proposed JAD module. The JAD loss L_{JAD} consists of the distance between the marginal distribution of $\widetilde{X}_u^m$ and $\widetilde{X}_v^h$, $\mathcal{D}_1(\widetilde{X}_u^m, \widetilde{X}_v^h)$ and the distance between the conditional distribution of $Y|\widetilde{X}_u^m$ and $Y|\widetilde{X}_v^h$, $\mathcal{D}_2(Y|\widetilde{X}_u^m, Y|\widetilde{X}_v^h)$. Specifically, $\widetilde{X}_u^m$ and $\widetilde{X}_v^h$ are the feature maps of the penultimate layer of the multimodal and hallucination networks, respectively. Y_u^m and Y_v^h are the logits output of multimodal and hallucination networks. $\phi_1(.)$ denotes the mapping function that transfers the $\widetilde{X}_u^m$ and $\widetilde{X}_v^h$ into high dimensional space. $\phi_2(.)$ denotes the mapping function that normalizes Y_u^m and Y_v^h into [0,1]. $E(.)$ denotes the expectation operator. L_{MSE} and L_{KL} denote the MSE and KL loss function, respectively.

layer and the logits output of the regular model pretrained with X_v, respectively.

C. Joint Adaptation Distillation

Different from conventional hallucination learning tasks, the modality-shared hallucination transfers knowledge from the multimodal network to the hallucination network, where both the input and structure of the teacher and student networks are different. This enlarges the representation discrepancy and would hinder knowledge transfer. To tackle this problem, we propose the JAD method that transfers the modality-shared knowledge from the multimodal network to the hallucination network by matching their joint probability distributions between the sample representation and groundtruth. This relaxes the constraints of matching feature values to avoid overfitting and retains the transfer probability between the sample representation and groundtruth to preserve comprehensive multimodal knowledge.

Fig. 2 illustrates the overall architecture of JAD, which is embedded into the penultimate layer and logits output layer of the multimodal and hallucination network in Fig. 1(a) and (b). It transfers the knowledge from the multimodal network to the hallucination network by minimizing the loss as follows:

$$L_{\text{JAD}} = \mathcal{D}_1\left(\widetilde{X}_u^m, \widetilde{X}_v^h\right) + \mathcal{D}_2\left(Y\middle|\widetilde{X}_u^m, Y\middle|\widetilde{X}_v^h\right) \tag{1}$$

where $\mathcal{D}_1(\widetilde{X}_u^m, \widetilde{X}_v^h)$ is the distance between the marginal distribution of $\widetilde{X}_u^m$ and $\widetilde{X}_v^h$, and $\mathcal{D}_2(Y|\widetilde{X}_u^m, Y|\widetilde{X}_v^h)$ is the distance between the conditional distribution of $Y|\widetilde{X}_u^m$ and $Y|\widetilde{X}_v^h$. Therefore, to calculate the $\mathcal{D}_1(\widetilde{X}_u^m, \widetilde{X}_v^h)$ and $\mathcal{D}_2(Y|\widetilde{X}_u^m, Y|\widetilde{X}_v^h)$, we need to estimate the distribution of $\widetilde{X}_u^m$, $\widetilde{X}_v^h$, $Y|\widetilde{X}_u^m$, and $Y|\widetilde{X}_v^h$, respectively. Although many works have been proposed to solve this problem, accurate distribution estimation is still difficult and complex [42], and this may limit the learning of the primary task. To tackle this problem, we make two innovations.

1) Inspired by the maximum mean discrepancy [44], we propose to approximate the $\mathcal{D}_1(\widetilde{X}_u^m, \widetilde{X}_v^h)$ with the maximum distance between the moments of $\widetilde{X}_u^m$ and $\widetilde{X}_v^h$. This avoids the problem of estimating their distributions and the $\mathcal{D}_1(\widetilde{X}_u^m, \widetilde{X}_v^h)$ can be rewritten as follows:

$$\mathcal{D}_1\left(\widetilde{X}_u^m, \widetilde{X}_v^h\right) = \sup_{\|\phi_1\|_{\mathcal{H}} \le 1} \left| E\left[\phi_1\left(\widetilde{X}_u^m\right)\right] - E\left[\phi_1\left(\widetilde{X}_v^h\right)\right] \right|^2 \tag{2}$$

where $\phi_1(.)$ means the mapping function that transforms $\widetilde{X}_u^m$, $\widetilde{X}_v^h$ into the space in arbitrary order, whose norm in the Hilbert space $\mathcal{H}$ is less than 1. $E[.]$ is the expectation operation.

Since the mean of the sample is the unbiased estimate of the expectation and $\|\phi_1\|_{\mathcal{H}} \le 1$, we can rewrite the $\mathcal{D}_1(\widetilde{X}_u^m, \widetilde{X}_v^h)$ as follows:

$$\mathcal{D}_1\left(\widetilde{X}_u^m, \widetilde{X}_v^h\right) = \left\| \frac{1}{b}\sum_{i=1}^{b} \phi_1\left(\widetilde{X}_u^i\right) - \frac{1}{b}\sum_{i=1}^{b} \phi_1\left(\widetilde{X}_v^i\right) \right\|_{\mathcal{H}}^2 \tag{3}$$

where $\widetilde{X}_u^i = (\widetilde{X}_u^m)_i$ and $\widetilde{X}_v^j = (\widetilde{X}_v^h)_j$ are the feature maps of the ith and jth samples in the X_u and X_v, respectively. Furthermore, this can be decomposed via the kernel method [45]

$$\mathcal{D}_1\left(\widetilde{X}_u^m, \widetilde{X}_v^h\right) = \frac{1}{b^2}\left\| \sum_{i,r} k\left(\widetilde{X}_u^i, \widetilde{X}_u^r\right) - 2\sum_{i,j} k\left(\widetilde{X}_u^i, \widetilde{X}_v^j\right) + \sum_{j,k} k\left(\widetilde{X}_v^j, \widetilde{X}_v^k\right) \right\| \tag{4}$$

where $k(x_1, x_2)$ is the kernel function. Similar to the previous study [44], we set it as the Gaussian function

$$k(x_1, x_2) = \exp\left(\frac{-\|x_1 - x_2\|^2}{\sigma}\right) \tag{5}$$

where exp(.) denotes exponential operation, σ denotes the variance. In this way, we can calculate the marginal distribution distance $\mathcal{D}_1(\widetilde{X}_u^m, \widetilde{X}_v^h)$ for specific $\widetilde{X}_u^m$ and $\widetilde{X}_v^h$.

2) We propose to estimate $\mathcal{D}_2(Y|\widetilde{X}_u^m, Y|\widetilde{X}_v^h)$ by the Monte Carlo algorithm. Specifically, the general formula for calculating $\mathcal{D}_2(Y|\widetilde{X}_u^m, Y|\widetilde{X}_v^h)$ is defined as follows:

$$\begin{aligned} \mathcal{D}_2\left(Y\middle|\widetilde{X}_u^m, Y\middle|\widetilde{X}_v^h\right) &= \text{KL}\left(P\left(Y\middle|\widetilde{X}_u^m\right)\middle\|Q\left(Y\middle|\widetilde{X}_v^h\right)\right) \\ &= \int P\left(Y\middle|\widetilde{X}_u^m\right)\log\frac{P\left(Y\middle|\widetilde{X}_u^m\right)}{Q\left(Y\middle|\widetilde{X}_v^h\right)} \end{aligned} \tag{6}$$

where KL(., .) denotes the KL-divergence. $P(Y|\widetilde{X}_u^m)$ and $Q(Y|\widetilde{X}_v^h)$ denote the distribution of $Y|\widetilde{X}_u^m$ and $Y|\widetilde{X}_v^h$, respectively. According to the Monte Carlo algorithm, we can convert the integral calculations to the sampling statistics problems [42], and $\mathcal{D}_2(Y|\widetilde{X}_u^m, Y|\widetilde{X}_v^h)$ can be

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rewritten as follows:

$$\mathcal{D}_2\left(Y|\widetilde{X}_u^b, Y|\widetilde{X}_v^h\right) = \sum_{i=1}^{b}\sum_{j=1,j\neq i}^{l} p_{j|i}\log\left(\frac{p_{j|i}}{q_{j|i}}\right) \quad (7)$$

where $p_{j|i}$ and $q_{j|i}$ are the samples drawn from the distribution of $P(Y|\widetilde{X}_u^b)$ and $Q(Y|\widetilde{X}_v^h)$, respectively. They denotes the conditional probability from the ith representations $(\widetilde{X}_u^m)_i$ and $(\widetilde{X}_v^h)_i$ to the jth category, respectively.

Thanks to the definition of neural network that the final logits output represents the likelihood that the sample belongs to each category [46], we can calculate the $p_{j|i}$ and $q_{j|i}$ through the Y_u^m and Y_v^h. Furthermore, to satisfy the basic properties of probability, we propose to normalize Y_u^m and Y_v^h into [0, 1] using the softmax function. Thus $p_{j|i}$ and $q_{j|i}$ can be rewritten as follows:

$$\begin{aligned} p_{j|i} &= \left[\phi_2\left(y_u^i\right)\right]_j \in [0,1] \\ q_{j|i} &= \left[\phi_2\left(y_v^i\right)\right]_j \in [0,1] \end{aligned} \quad (8)$$

where $y_u^i = (Y_u^m)_i$ and $y_v^i = (Y_v^h)_i$ denote the logits output of the ith sample of the X_u and X_v, respectively. $\phi_2(.)$ denotes the sotfmax function. $[.]_j$ means the jth component.

D. Modality-Shared Knowledge Transfer

The existing methods [13], [18], [20] usually handle the land cover classification with missing modalities by restoring the privilege modality information. On the one hand, this considers only the modality-specific properties and ignores the intermediate cross-modal interaction that is crucial for multimodal classification. On the other hand, this does not scale well with the number of missing modalities, since it needs to train the privileged branch for each missing modality. To tackle these problems, we propose the modality-shared hallucination method to guide the hallucination model to learn the modality-shared knowledge from the multimodal model trained with complete data directly.

As shown in Fig. 1, we take the common setting that $\mathcal{U} = 2$ and $\mathcal{V} = 1$ as an example to explain the mechanism of the modality-shared knowledge transfer. Here we suppose M_2 as the privilege modality. Overall, we propose to guide the hallucination network to learn the shared knowledge among modalities $\{M_1, M_2\}$ from the pretrained multimodal network. Specifically, the training of multimodal models has been extensively studied, and numerous efficient fusion algorithms, such as cross fusion [47], channel selection [7], and cross-channel reconstruction [5], have been proposed. Since the multimodal model training is independent of the knowledge transfer, here it can be implemented with various structures and fusing strategies. Besides, because the effectiveness of knowledge transfer is highly dependent on structural similarity [27], [28], the architecture of the hallucination network is determined by removing the modules that are used only by the missing modality in the multimodal network, so that the structural discrepancy between them is minimized. Furthermore, to cope with the remaining structural and input differences, the JAD method in Fig. 2 is introduced to transfer the modality-shared knowledge from the multimodal network to the hallucination network by matching their joint distributions between the sample representations and groundtruths. Finally, the total loss of stage one is defined as follows:

$$L_{\mathrm{MSKT}} = L_{\mathrm{CE}}(X_v, Y) + \alpha\mathcal{D}_1\left(\widetilde{X}_u^m, \widetilde{X}_v^h\right) + \beta\mathcal{D}_2\left(Y\middle|\widetilde{X}_u^m, Y\middle|\widetilde{X}_v^h\right) \quad (9)$$

where $L_{\mathrm{CE}}(X_v, Y)$ is label supervision loss for the task at hand. α and β are the balance hyper-parameter.

E. Shared and Specific Information Fusion

Although the modality-shared knowledge transfer alleviates the information loss of missing $\{M_2\}$ by guiding the hallucination network to restore the shared knowledge among the modalities$\{M_1, M_2\}$ from the incomplete modality $\{M_1\}$, it considers only the shared knowledge and ignores the specific information of the modality $\{M_1\}$. Note that the modality-specific information is also of great importance in the multimodal remote sensing land cover classification [10]. To solve this problem, we introduce the shared and specific information fusion to take the regular model trained with the modalities $\{M_1\}$ into consideration.

As shown in Fig. 1, we propose a two-stream network to weigh the prediction results produced by hallucination and regular models to make the final decision so that the inference model can leverage the modality-shared and specific information. The final classification result of unlabeled data x_v is defined as follows:

$$R(x_v) = (1-\eta)\, f_h(\varphi_h(x_v)) + \eta f_n(\varphi_n(x_v)) \quad (10)$$

where η is a learnable weighting hyper-parameter. $f_h(.)$ and $f_n(.)$ are the nonlinear mapping function between the sample representation and the logits output, of the hallucination and regular branches, respectively. Furthermore, to achieve better performance, we freeze the parameters of feature encoders of both branches and fine-tune their fully-connected layer on the dataset Φ by the following training loss:

$$L_{\mathrm{MS2F}} = L_{\mathrm{CE}}(R(X_v), Y) \quad (11)$$

where $L_{\mathrm{CE}}(.,.)$ is the cross-entropy loss.

F. Training

The training of the proposed MSH-Net consists of three steps: baseline training, hallucination training, and ensemble training.

1) Baseline Training: The baseline training aims to train the multimodal network with complete data modality in Θ and the regular network with the incomplete data modality in Ψ for modality-shared knowledge transfer as well as shared and specific information fusion, respectively. Specifically, the architecture of the multimodal network is set as the MDL-CNNs-Net [47], which has been proven to be effective in multimodal land cover classification. It is worth noting that the proposed MSH-Net is model-agnostic and the multimodal network can also be implemented with other architectures, such as CCR-Net [5] and CHGFNet [48]. As shown in Fig. 1,

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the architecture of the regular network is the same as that of the hallucination network, which would be described in the hallucination training in detail. Finally, the training loss for both multimodal and regular networks is the cross-entropy loss.

2) Hallucination Training: As shown in Fig. 1, the hallucination training aims to train the hallucination network in the stage of modality-shared knowledge transfer. The architecture of the hallucination network is determined by the procedure introduced in Section III-D, so that the structure differences between multimodal and hallucination networks are minimized. Specifically, as shown in Fig. 1, since M_2 is the privilege modality, the feature extractor used only by M_2 in the multimodal network is removed, and the remaining one used by M_1 is set as the feature extractor of the hallucination network. Then the feature fusion and FC modules used by M_1 are also preserved for the hallucination network. Here, since the input of the hallucination network is only M_1, we name the feature fusion module as the feature transformer. The training loss is the L_{MSKT} proposed in (9). Finally, it is worth noting that the parameters of the pretrained multimodal network are frozen and only the hallucination network is learnable.

3) Ensemble Training: As shown in Fig. 1, this step trains the ensemble network that weighs the logits output of hallucination and regular networks to make the final decision. The training loss is the L_{MS2F} proposed in (11). Specifically, the parameters of their feature encoders and feature transformer module are frozen and only the FC module and weighting factor η are learnable.

IV. Experiments

A. Datasets

In the experiments, two multimodal datasets, Huston2013 and Augsburg, are used for performance assessment quantitatively and qualitatively. A brief description of the two datasets is given as follows.

1) Huston2013: This dataset consists of three different data sources, including a hyperspectral (HS) image, an MS image, and a LiDAR image. Among them, the HS and LiDAR images are available from the IEEE GRSS data fusion contest 2013 [8]. The MS image is generated by degrading the original HS image in spatial and spectral domains [10]. Specifically, the HS image is composed of 144 HS bands with the size 349 × 1905 and spatial resolution of 2.5 m. The resulting MS image is composed of the same size as the original HS image and eight spectral bands at a ground sampling distance (GSD) of 2.5 m. The LiDAR image is a grayscale image of the same spatial size as the HS image. Finally, a total of 15 029 groundtruth samples over 15 classes are available, as detailed in Table I.

2) Augsburg: This dataset consists of three different data sources, including an HS image, a dual-polarized SAR image, and a DSM image [10]. Specifically, the HS image is composed of 180 bands. The SAR image consists of four feature channels. The DSM image is a grayscale image. The dimension of each band is 332 × 485 pixels. Finally, as shown in Table II, there are a total of seven classes in the image, groundtruth of which is available for 78 294 pixels that are divided into 761 training pixels and 77 533 test pixels.

TABLE I
Numbers of Training and Testing Samples for Each Class in the Huston Dataset

No.	Ground Object Name	Training Set	Testing Set
1	Healthy Grass	198	1053
2	Stressed Grass	190	1064
3	Synthetic Grass	192	505
4	Tree	188	1056
5	Soil	186	1056
6	Water	182	143
7	Residential	196	1072
8	Commercial	191	1053
9	Road	193	1059
10	Highway	191	1036
11	Railway	181	1054
12	Parking Lot	192	1041
13	Parking Lot2	184	285
14	Tennis Court	181	247
15	Running Track	187	473
	Total	2832	12197

TABLE II
Numbers of Training and Testing Samples for Each Class in the Augsburg Dataset

No.	Ground Object Name	Training Set	Testing Set
1	Forest	146	13361
2	Residential Area	264	30065
3	Industrial Area	21	3830
4	Low Plants	248	26609
5	Allotment	52	523
6	Commercial Area	7	1638
7	Water	23	1507
	Total	761	77533

B. Experiments Setup

1) Evaluation Metrics: In the experiments, we choose the multimodal land cover classification task to evaluate the proposed framework. This has been regarded as a potential application to evaluate the quality of multimodal learning [10]. Specifically, we report three metrics from common land cover classification evaluations, including the overall accuracy (OA), average accuracy (AA), and Kappa Coefficient (Ka) for quantitative performance comparison.

2) Implementation Details: The proposed method is implemented on the PyTorch platform. The entire framework is trained by using SGD optimizer with the training epoch 500, batch size 32, learning rate 1e-3, and momentum 0.9. Moreover, for all the models, we apply the classic cosine learning rate policy to update the learning rate of each iteration, so that they can achieve better performance. Finally, to take the randomness in initialization into consideration, the averaged results will be reported out of three runs.

In addition to the training setting, the data preparation is also of great importance to the performance. Different from the conventional classification task that takes the whole image as the input of the network, the remote sensing image

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classification needs to crop the patch centered by a pixel as the input. For a fair comparison, we crop the patch in the same way as the previous work [47] by extending the original pixel via the "replicate" operation and setting the patch size as 7×7. Besides, we use random flipping for data augmentation.

3) Modality Combinations: To evaluate the generalization of our proposed MSH-Net, we perform experiments on four common multimodal combinations. In addition to the combinations of HS and MS images (HS-MS), we also considered the highly heterogeneous combinations, including HS and LiDAR images (HS-LiDAR), and HS and SAR images (HS-SAR). Besides, to evaluate the scalability of our proposed MSH-Net in the scenarios where multiple modalities (more than one) are missing, we also perform experiments on the combinations of HS, SAR, and DSM images (HS-SAR-DSM).

4) Comparison Methods: To evaluate the effectiveness of privilege modality learning and the proposed MSH-Net on handling the land cover classification with missing modalities, we compare the MSH-Net with two categories of methods.

1) The first category reports performance of the multimodal and regular models that are trained and tested with the same modality data. Specifically, the multimodal model is trained with complete data modalities of each modality combination while the regular model is trained only with partial modalities that are not missing during the inference. For a simplified description, the name of these models of each modality combination is referred to as the name of the modality they used. Take the combination of HS and MS images as an example, the *HS-Net* means the model trained with HS image; the *MS-Net* means the model trained with MS image; the *HS-MS-Net* means the model trained with HS and MS images simultaneously.
2) The second category reports the performance of the state-of-the-art privileged modality learning methods that transfer the privileged knowledge by different distillation methods, including the Hall-Net [18] by FD, the MARS-Net [26] by logit-based distillation (LD), the ADD-Net [23] by adversarial discriminative distillation (ADD), and the DH-ADNet [13] by DHAD.

C. Configuration of Hyperparameters

As discussed above, the proposed distillation strategies have two hyper-parameters α and β, which would influence the land cover classification performance of our proposed method. To explore the impact of the two parameters, we conduct experiments for the modality combination of HS-MS, HS-LiDAR, and HS-SAR by varying α from 1.0 to 5.0 with a span of 1.0, and the values of the parameters β ranging from 1.0 to 5.0 with a span of 1.0 and keeping the other parameters fixed. The result is shown in Fig. 3, from which we can get the optimal parameter combination for different modality combinations.

Firstly, for the combination of HS-MS, as shown in Fig. 3(a) and (b), it is better to set (α, β) as $(1, 2)$ when the HS modality is missing and $(1, 1)$ when the MS modality is missing. Besides, for the combination of HS-LiDAR, as shown in Fig. 3(c) and (d), when the HS modality is missing, $(1, 5)$ is a better choice for (α, β) and when LiDAR modality is missing, $(2, 1)$ is better. Furthermore, for the combination of HS-SAR, as shown in Fig. 3(e) and (f), we can set (α, β) as $(1, 5)$ when the HS modality is missing, and $(2, 1)$ when the SAR modality is missing, so that the model can reach the highest accuracy, respectively.

In particular, we can see that the optimal hyperparameter (α, β) for the modality combination of HS-LiDAR and HS-SAR is the same $(1, 5)$ when only the LiDAR or SAR modality is available for inference (HS modality is missing), and $(2, 1)$ when only the HS modality is available (LiDAR or SAR modalities is missing). The reason may be that both of them are highly heterogeneous combinations. Thus for the modality combination of HS-SAR-DSM, we also set the (α, β) as $(1, 5)$ when only the SAR or DSM modality is available for inference and $(2, 1)$ when only the HS modality is available.

TABLE III

PERFORMANCE OF DIFFERENT METHODS ON THE HUSTON2013 DATASET WITH THE HS AND MS MODALITIES. "BASELINE" MEANS THAT THERE IS NO MISSING MODALITY DURING THE TEST. "W/O" MEANS MISSING CORRESPONDING MODALITIES DURING THE TEST. "HS-NET" AND "MS-NET" MEAN THE MODELS TRAINED WITH HS AND MS MODALITIES, RESPECTIVELY. HS-MS-NET MEANS THE MODEL TRAINED WITH HS AND MS MODALITIES SIMULTANEOUSLY

Method	Training Modalities	Testing Modalities	OA (%)	AA (%)	Ka
Baseline					
HS-Net	HS	HS	84.58	86.66	83.39
MS-Net	MS	MS	86.11	88.88	84.92
HS-MS-Net	HS, MS	HS, MS	**87.54**	**90.01**	**86.51**
W/o HS modality					
HS-MS-Net	HS, MS	MS	60.05	61.23	58.18
Hall-Net	HS, MS	MS	86.41	88.56	85.21
MARS-Net	HS, MS	MS	86.42	88.82	85.25
ADD-Net	HS, MS	MS	86.71	89.01	85.11
DH-ADNet	HS, MS	MS	86.85	89.21	85.36
TAKD-Net	HS, MS	MS	86.66	88.93	85.23
DASK-Net	HS, MS	MS	86.78	89.07	85.29
MSH-Net	HS, MS	MS	**87.12**	**89.70**	**86.02**
W/o MS modality					
HS-MS-Net	HS, MS	HS	72.87	73.56	69.23
Hall-Net	HS, MS	HS	85.84	87.65	84.33
MARS-Net	HS, MS	HS	85.90	87.81	84.73
ADD-Net	HS, MS	HS	86.21	87.34	85.12
DH-ADNet	HS, MS	HS	86.56	87.69	85.48
TAKD-Net	HS, MS	HS	86.37	87.54	85.32
DASK-Net	HS, MS	HS	86.49	87.88	85.68
MSH-Net	HS, MS	HS	**87.25**	**88.63**	**86.17**

D. Result and Analysis on HS-MS Combination

Table III lists the quantitative performance comparison in terms of OA, AA, and Ka in three different experimental settings on the HS-MS-LiDAR dataset, i.e., baseline, w/o HS modality, and w/o MS modality, respectively.

First, from the setting of complete modality, we can observe that the HS-MS-Net trained with HS and MS modalities data outperforms the HS-Net and MS-Net that are trained only

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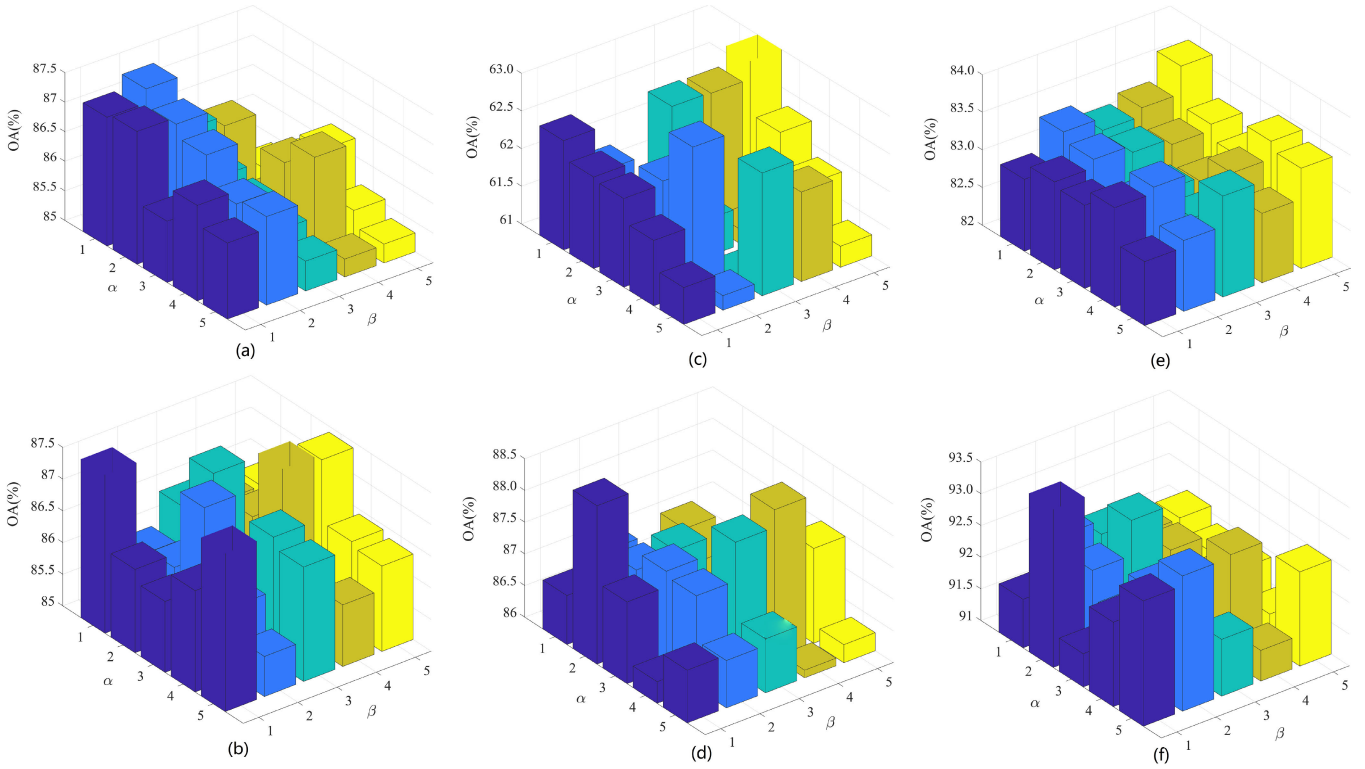


Fig. 3. Depiction of the trend between the OA and the parameters α, β on three modality combinations. (a) and (b) OA of the combination of HS-MS when MS or HS modality is missing. (c) and (d) OA of the combination of HS-LiDAR when LiDAR or HS modality is missing. (e) and (f) OA of the combination of HS-SAR when SAR or HS modality is missing.

with HS modality data and MS modality data, respectively. This shows the effectiveness of multimodal learning for land cover classification. However, the OA of the HS-MS-Net drops by 27.49% and 14.38% when the HS modality and MS modality are missing, respectively. This indicates that it is not effective to apply the conventional multimodal model to the practical application where the testing modalities are incomplete. In contrast, the proposed MSH method overcomes this issue and retains the performance gain compared to the unimodal model. Specifically, the OA, AA, and Ka of the MSH method outperform the MS-net by 1.01%, 0.82% and, 1.12%, and the HS-net by 2.67%, 1.97% and, 2.78%, respectively.

More significantly, the resulting model trained by the MSH method is capable of transferring the modality-shared information from the multimodal model to the hallucination model effectively, yielding higher performance over the distillation-based methods that only preserve the modality-specific knowledge. Specifically, when the HS modality is missing, the hallucination stream distills the shared knowledge among the HS and MS modality from the multimodal model to the hallucination model trained with the MS modality via the procedure in Fig. 1(a). This outperforms the conventional distillation-based methods, including DH-ADNet, ADD-Net, MARS-Net, and Hall-Net by 0.27%, 0.41%, 0.7%, and 1.28% in the OA. When the MS modality is missing, the MSH method outperforms them by 0.73%, 1.04%, 1.35%, and 1.41% in the OA.

E. Result and Analysis on HS-LiDAR Combination

This section further evaluates the performance of the proposed MSH method in the scenarios where HS or LiDAR modality is missing, to study its generalization to the highly heterogeneous modality combination. Table IV reports the quantitative performance comparison in terms of OA, AA, and Ka in three different experimental settings on the HS-MS-LiDAR dataset, i.e., baseline, w/o HS modality, and w/o LiDAR modality, respectively.

From Table IV, we can see that when the HS modality is missing and only the LiDAR modality is available for inference, the OA of HS-LiDAR-Net drops by 79.53% and is 45.08% worse than that of the LiDAR-Net. On the other hand, when the LiDAR modality is missing and only the HS modality is available for inference, the OA of HS-LiDAR-Net drops by 44.88% and is 37.48% worse than that of the HS-Net. Compared to the combination of HS and MS modalities, the performance degradation caused by the missing modality is more significant here. Thus, it is of great importance to address the problem of deploying the model trained with highly heterogeneous modalities data in practical scenarios where partial modalities are missing.

Besides, it is worth noting that the MSH-Net also performs the best for all three indicators over the model trained by unimodal modality. Specifically, it improves the OA, AA, and Ka of the LiDAR-Net by 5.1%, 5.19%, and 5.46%, respectively, when only the LiDAR modality is available for inference, and HS-Net by 3.42%, 2.73%, and 3.60%, respectively, when only

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TABLE IV

PERFORMANCE OF DIFFERENT METHODS ON THE HUSTON2013 DATASET WITH HS AND LIDAR MODALITIES. "LIDAR-NET" IS THE MODEL TRAINED WITH LIDAR MODALITY. HS-LIDAR-NET MEANS THE MODEL TRAINED WITH HS AND LIDAR MODALITIES SIMULTANEOUSLY

Method	Training Modalities	Testing Modalities	OA(%)	AA(%)	Ka
Baseline					
HS-Net	HS	HS	84.58	86.66	83.39
LiDAR-Net	LiDAR	LiDAR	57.53	58.21	54.07
HS-LiDAR-Net	HS, LiDAR	HS, LiDAR	**91.98**	**92.42**	**91.29**
W/o HS modality					
HS-LiDAR-Net	HS, LiDAR	LiDAR	12.45	13.54	9.81
Hall-Net	HS, LiDAR	LiDAR	59.68	60.97	56.43
MARS-Net	HS, LiDAR	LiDAR	59.71	60.98	56.45
ADD-Net	HS, LiDAR	LiDAR	60.12	61.01	57.09
DH-ADNet	HS, LiDAR	LiDAR	60.98	61.11	57.75
TAKD-Net	HS, LiDAR	LiDAR	60.33	61.06	57.34
DASK-Net	HS, LiDAR	LiDAR	61.34	61.98	58.02
MSH-Net	HS, LiDAR	LiDAR	**62.63**	**63.64**	**59.53**
W/o LiDAR modality					
HS-LiDAR-Net	HS, LiDAR	HS	47.10	49.22	46.51
Hall-Net	HS, LiDAR	HS	85.57	86.83	84.44
MARS-Net	HS, LiDAR	HS	85.66	86.94	84.60
ADD-Net	HS, LiDAR	HS	85.24	87.46	85.01
DH-ADNet	HS, LiDAR	HS	86.57	88.02	85.45
TAKD-Net	HS, LiDAR	HS	85.76	87.64	85.12
DASK-Net	HS, LiDAR	HS	86.89	88.31	85.82
MSH-Net	HS, LiDAR	HS	**88.00**	**89.39**	**86.99**

TABLE V

PERFORMANCE OF DIFFERENT METHODS ON THE AUGSBURG DATASET WITH HS AND SAR MODALITIES. "SAR-NET" IS THE MODEL TRAINED WITH SAR MODALITY. HS-SAR-NET MEANS THE MODEL TRAINED WITH HS AND SAR MODALITIES SIMULTANEOUSLY

Method	Training Modalities	Testing Modalities	OA(%)	AA(%)	Ka
Baseline					
HS-Net	HS	HS	90.91	65.32	86.89
SAR-Net	SAR	SAR	82.68	46.12	74.75
HS-SAR-Net	HS, SAR	HS, SAR	**92.18**	**63.74**	**88.59**
W/o HS modality					
HS-SAR-Net	HS, SAR	SAR	1.98	14.13	0.04
Hall-Net	HS, SAR	SAR	83.08	**46.02**	75.27
MARS-Net	HS, SAR	SAR	83.11	46.01	75.29
ADD-Net	HS, SAR	SAR	83.22	45.25	75.35
DH-ADNet	HS, SAR	SAR	83.34	45.20	75.46
TAKD-Net	HS, SAR	SAR	83.28	45.23	75.39
DASK-Net	HS, SAR	SAR	83.45	45.21	75.65
MSH-Net	HS, SAR	SAR	**83.70**	45.18	**75.97**
W/o SAR modality					
HS-SAR-Net	HS, SAR	HS	44.81	26.59	11.81
Hall-Net	HS, SAR	HS	90.97	64.09	86.92
MARS-Net	HS, SAR	HS	91.02	64.11	86.98
ADD-Net	HS, SAR	HS	91.56	64.76	87.52
DH-ADNet	HS, SAR	HS	91.88	65.69	88.23
TAKD-Net	HS, SAR	HS	91.72	65.12	87.88
DASK-Net	HS, SAR	HS	92.12	65.91	88.81
MSH-Net	HS, SAR	HS	**93.27**	**67.08**	**90.26**

the HS modality is available for inference. The above improvement demonstrates the generalization of MSH-Net to deal with the remote sensing classification with missing modalities in highly heterogeneous modality combinations. Furthermore, the MSH-Net outperforms the DH-ADNet, ADD-Net, MARS-Net, and Hall-Net by 1.74%, 2.51%, 2.92%, and 2.95% in the OA, respectively, when the HS modality is missing and by 1.43%, 1.76%, 2.34%, and 2.43%, respectively, when the LiDAR modality is missing. This demonstrates that it is still effective and superior to handle the missing modality by transferring the modality-shared knowledge to the hallucination in scenarios where multiple modalities are highly heterogeneous.

Finally, Fig. 4 shows the classification maps of considered compared algorithms in the given scene. There is an identical trend in both quantitative and qualitative comparisons (between Fig. 4 and Table IV) and the classification map derived from the proposed MSH-Net is better than other comparative methods. Taking the commercial area as an example, in the black box, it can be seen that the commercial classification map derived from our MSH-Net is more complete and has fewer misclassified. However, the HS-LiDAR-Net is even not capable of extracting the commercial area. The commercial classification maps derived from the HS-Net, Hall-Net, MARS-Net, and ADD-Net are intermittent and the commercial classification map derived from DH-ADNet has some misclassification.

F. Result and Analysis on HS-SAR Combination

We also perform comparison experiments on the combination of HS and SAR modality of the Augsburg dataset to further evaluate the generalization of the MSH-Net. Specifically, the HS and SAR data are also highly heterogeneous due to different imaging mechanisms. Table IV reports the quantitative performance comparison in terms of OA, AA, and Ka in three different experimental settings, i.e., baseline, w/o HS modality, and w/o SAR modality, respectively.

As shown in Table V, the proposed MSH-Net has a better performance than the conventional multimodal model and overcomes other competitors when partial modality is missing. Specifically, for the situation where the HS modality is missing, the OA of the MSH-Net outperforms that of the DH-ADNet, ADD-Net, MARS-Net, Hall-Net, and HS-SAR-Net by 0.36%, 0.48%, 0.59%, 0.62%, and 71.27%, respectively. For the situation where the SAR modality is missing, the OA of the MSH-Net outperforms that of the DH-ADNet, ADD-Net, MARS-Net, Hall-Net, and HS-SAR-Net by 1.39%, 1.71%, 2.25%, 2.45%, and 48.46%, respectively. These results again demonstrate the effectiveness to overcome the modality missing problem of the highly heterogeneous modality combination by guiding the hallucination model to learn from the multimodal model instead of the privilege model.

Furthermore, the MSH-Net also improves the OA, AA, and Ka of the HS-Net by 2.36%, 1.76%, and 3.37%, when only the HS modality is available for inference. This demonstrates the validity to introduce extra modality-shared knowledge. More significantly, it even outperforms the HS-SAR-Net that is trained and tested with complete modalities by 1.17%, 3.34%, and 1.67% in terms of AA, OA, and Ka, respectively. This is probably because we utilize both modality-shared and

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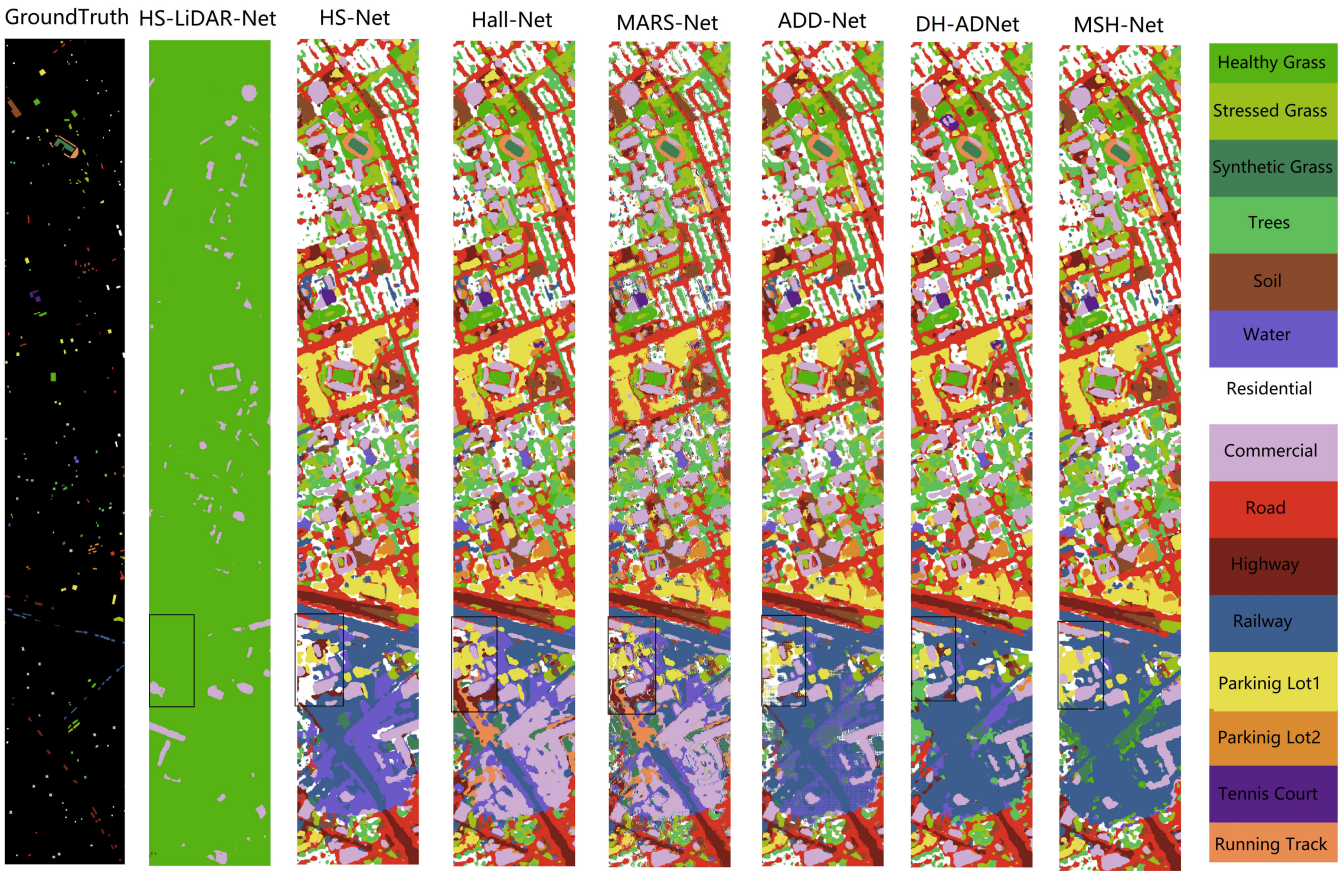


Fig. 4. Classification maps of different compared algorithms under the setting of "w/o LiDAR modality" of the HS-LiDAR combination of the Huston2013 dataset.

modality-specific features while HS-SAR-Net only considers modality-shared features between HS and SAR images. This shows the potential of our method in the multimodal remote sensing classification, to deal with the challenge of missing modality while surpassing multimodal models trained with complete modalities. Finally, although the MSH-Net performs better than the SAR-Net in the OA and Ka when only the SAR modality is available for inference, its AA is worse than that of the SAR-Net and other competitors. In particular, we can see that this phenomenon also exists in the baseline setting, where the OA of the HS-SAR-Net is 1.27 better than that of the HS-Net while the AA is 1.58% worse than that of the HS-Net. The reason behind this may be the imbalanced sample distribution that guides the model to improve the OA while ignoring the accuracy of each category.

Finally, Fig. 5 shows the classification maps. It is obvious that the classification map of our method is better than other contrast methods. Such as, for the residential area and water in the black box, it can be seen that the residential area water classification map derived from our MSH-Net is more complete and has fewer misclassified. In contrast, the classification maps derived from other methods are confused by the industrial area.

G. Result and Analysis on HS-SAR-DSM Combination

This section performs experiments on the modality combination of the HS, SAR, and DSM images to evaluate the scalability of the MSH-Net in the scene with multiple missing modalities. Specifically, three are four settings, i.e., baseline, w/o HS and SAR modalities, w/o SAR and DSM modalities, and w/o HS and DSM modalities, respectively. Table VI reports the quantitative performance comparison in terms of OA, AA, and Ka for each setting.

As shown in Table VI, we can see that the proposed MSH-Net performs better than the conventional multimodal model under the setting of missing modalities, and the unimodal model under the baseline setting. Specifically, compared to the HS-SAR-DSM-Net, the MSH-Net increases the OA from 1.98% to 81.86% when HS and SAR modalities are missing, from 3.3% to 83.35% when HS and DSM modalities are missing, and from 72.03% to 92.99% when SAR and DSM modalities are missing. These results demonstrate the effectiveness of the MSH-Net to overcome the issues of missing multiple modalities. Furthermore, the MSH-Net outperforms the baseline methods, HS-Net by 1.98% in OA, SAR-Net by 0.85%, and the DSM-Net by 3.45%. These results demonstrate that our proposed modal sharing feature transfer is still effective when multiple modalities are missing.

H. Ablation Study

In this section, we will study the effectiveness of the important design elements in the proposed method, including the JAD and modality-shared knowledge transfer. To this end, we conduct extensive ablation experiments on the modality

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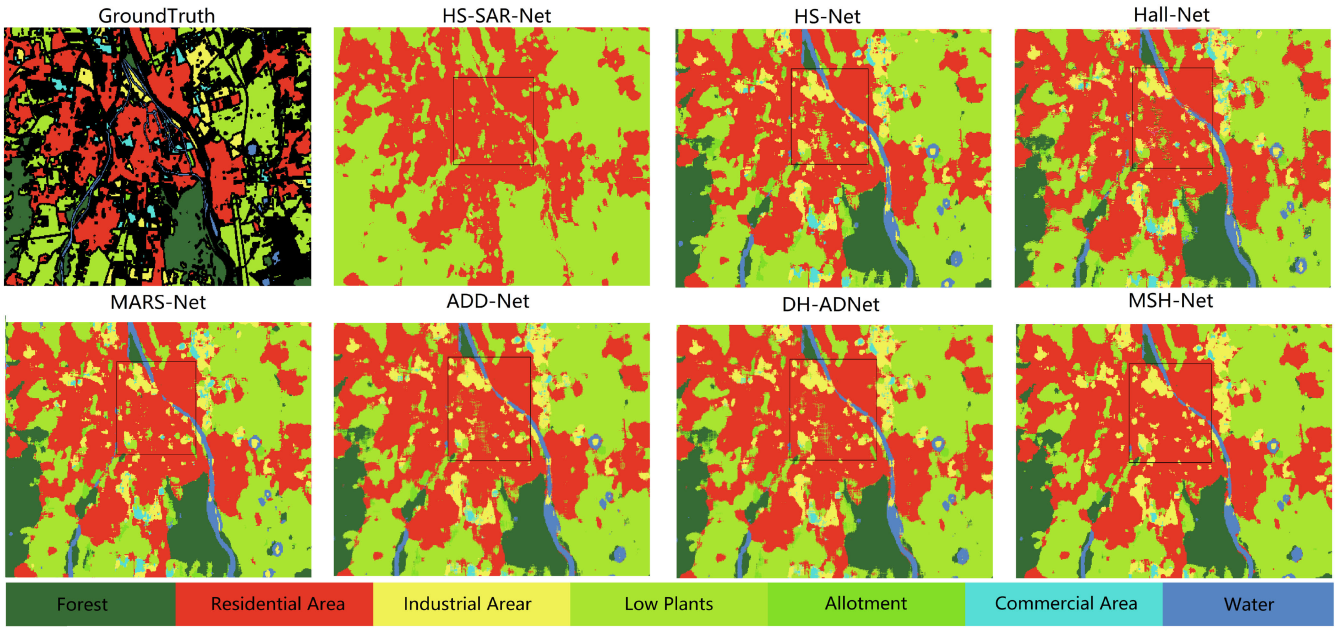


Fig. 5. Classification maps of different compared algorithms under the setting of "w/o SAR modality" of the HS-SAR combination of the Augsburg dataset.

TABLE VI

PERFORMANCE OF DIFFERENT METHODS ON THE AUGSBURG DATASET WITH HS, SAR, AND DSM MODALITIES. "DSM-NET" IS THE MODEL TRAINED WITH DSM MODALITY. "HS-SAR-DSM-NET" MEANS THE MODEL TRAINED WITH HS, SAR, AND DSM MODALITIES SIMULTANEOUSLY

Method	Training Modalities	Testing Modalities	OA(%)	AA(%)	Ka
Baseline					
HS-Net	HS	HS	90.91	65.32	86.89
SAR-Net	SAR	SAR	82.68	46.12	74.75
DSM-Net	DSM	DSM	78.12	41.96	67.64
HS-SAR-DSM-Net	HS, SAR,DSM	HS,SAR,DSM	**93.31**	**63.85**	**90.25**
W/o HS and SAR modalities					
HS-SAR-DSM-Net	HS, SAR,DSM	DSM	1.98	14.28	0
MSH-Net	HS, SAR,DSM	DSM	**81.61**	**43.60**	**72.50**
W/o HS and DSM modalities					
HS-SAR-DSM-Net	HS, SAR,DSM	SAR	3.3	17.92	1.28
MSH-Net	HS, SAR,DSM	SAR	**83.51**	**44.11**	**75.57**
W/o SAR and DSM modalities					
HS-SAR-DSM-Net	HS, SAR,DSM	HS	72.03	40.32	57.42
MSH-Net	HS, SAR,DSM	HS	**92.99**	**66.86**	**89.85**

combination of HS-LiDAR with different settings for the MSH-Net.

1) Effect of JAD Strategy: To study the effect of the JAD strategy, we report and compare the performance of the MSH-Net implemented with different privilege distillation strategies. Specifically, the comparison distillation methods include the FD and DHAD that match the value of feature maps, the LD that matches the distribution of logit output and the ADD that matches the distribution of feature maps. As shown in Table VII, the MSH-Net implemented with the JAD achieves the best performance under the setting of w/o HS modality and w/o LiDAR modality, respectively. This demonstrates the superiority of transferring the modality-shared knowledge by matching the joint distribution probability between the sample representation and groundtruth of the multimodal model and the hallucination model. Specifically, the MSH-Net implemented with ADD method outperforms those implemented with DHAD and FD methods by 0.37% and 2.97% under the setting of w/o HS modality, showing the effectiveness of matching the relation instead of the value to overcome the representation discrepancy between multimodal and hallucination network. Moreover, the MSH-Net implemented with the JAD method outperforms that implemented with the ADD method by 0.98%, verifying the effectiveness of further introducing the conditional constraints between the sample representation and groundtruth for the relation-based distillation method.

2) Effect of Modality-Shared Knowledge Transfer: Here we report and compare the performance of the MSH-Net and conventional modality hallucination network (MH-Net)

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TABLE VII

PERFORMANCE OF THE MSH-NETS THAT ARE IMPLEMENTED WITH DIFFERENT PRIVILEGE DISTILLATION STRATEGIES ON THE HUSTON2013 DATASET WITH HS AND LIDAR MODALITIES

Method	Training Modalities	Testing Modalities	OA(%)	AA(%)	Ka
W/o HS modality					
FD	HS, LiDAR	LiDAR	58.68	59.97	55.43
LD	HS, LiDAR	LiDAR	60.91	61.68	57.98
ADD	HS, LiDAR	LiDAR	61.65	62.33	58.09
DHAD	HS, LiDAR	LiDAR	60.18	61.11	57.75
JAD	HS, LiDAR	LiDAR	**62.63**	**63.64**	**59.53**
W/o LiDAR modality					
FD	HS, LiDAR	HS	87.57	88.53	86.24
LD	HS, LiDAR	HS	86.66	87.94	85.60
ADD	HS, LiDAR	HS	87.67	88.83	85.65
DHAD	HS, LiDAR	HS	87.14	88.16	86.51
JAD	HS, LiDAR	HS	**88.00**	**89.39**	**86.99**

TABLE VIII

PERFORMANCE OF THE PROPOSED MSH-NET AND THE CONVENTIONAL MH-NET IMPLEMENTED WITH THE JAD METHOD ON THE HUSTON2013 DATASET WITH HS AND LIDAR MODALITIES

Method	Training Modalities	Testing Modalities	OA(%)	AA(%)	Ka
W/o HS modality					
MH-Net	HS, LiDAR	LiDAR	60.95	62.12	57.65
MSH-Net	HS, LiDAR	LiDAR	**62.63**	**63.64**	**59.53**
W/o LiDAR modality					
MH-Net	HS, LiDAR	HS	86.99	87.83	85.85
MSH-Net	HS, LiDAR	HS	**88.00**	**89.39**	**86.99**

implemented with the JAD method to study the effect of transferring the modality-shared knowledge and privilege modality knowledge. As shown in Table VIII, the MSH-Net outperforms the MH framework by 1.68% and 1.01% in the OA under the setting of w/o HS modality and w/o LiDAR modality, respectively. This demonstrates the superiority of transferring the modality-shared knowledge and privilege modality from the multimodal model to the hallucination model.

3) Effect of Backbone Network: The proposed MSH-Net is model-independent, and the multimodal network can be implemented with other architectures besides MDL-CNNs-Net. Following the reviewer's suggestion, we implement the MSH-Net with two different backbone networks, CCR-Net [5] and PC2ANet [49]. For fair compassion with the baseline method, we conduct experiments on the HS-LiDAR combination in the Huston2013 dataset which is also considered by CCR-Net and PC2ANet.

As shown in Table IX, for the setting of "W/ HS," MSH-Net achieves the best performance regardless of using CCR-Net or PC2ANet as the backbone. Particularly, MSH-Net increases the performance of the HS-Net with the same backbone network by 3.12% and 2.72% in OA, respectively. Besides, for the setting of "W/ LiDAR," MSH-Net also achieves the best performance regardless of using CCR-Net or PC2ANet as the backbone. It increases the performance of LiDAR-Net with the same backbone network by 4.58% and 4.77% in OA, respectively. These results demonstrate that the proposed MSH-Net is robust for different backbone networks.

TABLE IX

PERFORMANCE OF THE MSH-NETS THAT ARE IMPLEMENTED WITH DIFFERENT BACKBONE ON THE HUSTON2013 DATASET WITH HS AND LIDAR MODALITIES. "W/" MEANS USING CORRESPONDING MODALITIES FOR THE TEST

Backbone	Method	Training Modalities	OA(%)	AA(%)	Ka
W/ HS					
CCR-Net	HS-Net	HS	82.34	84.67	81.43
	HS-LiDAR-Net	HS,LiDAR	13.39	13.13	5.79
	MSH-Net	HS,LiDAR	**85.46**	**87.19**	**84.52**
PC2ANet	HS-Net	HS	91.16	91.73	89.12
	HS-LiDAR-Net	HS,LiDAR	74.22	75.89	72.14
	MSH-Net	HS,LiDAR	**93.88**	**94.19**	**92.12**
W/ LiDAR					
CCR-Net	LiDAR-Net	LiDAR	51.29	52.73	46.92
	HS-LiDAR-Net	HS,LiDAR	13.67	14.13	8.17
	MSH-Net	HS,LiDAR	**55.87**	**56.91**	**52.66**
PC2ANet	LiDAR-Net	LiDAR	59.89	61.21	56.92
	HS-LiDAR-Net	HS,LiDAR	23.78	25.44	19.10
	MSH-Net	HS,LiDAR	**64.66**	**65.78**	**61.02**

TABLE X

COMPARISON OF THE PARAMETER AND TRAINING FLOPS OF MODELS. ALL NUMBERS ARE PRESENT WITH THE UNIT M (=1E6)

Method	Privilege		Hallucination		All	
	Params	FLOPs	Params	FLOPs	Params	FLOPs
Hall-Net	0.508	20.72	0.127	5.43	0.636	26.15
MARS-Net	0.508	20.72	0.127	5.43	0.636	26.15
ADD-Net	0.508	20.72	0.198	7.33	0.707	28.05
DH-ADNet	0.508	20.72	0.237	8.54	0.746	29.26
TAKD-Net	0.508	20.72	0.254	10.86	0.762	31.58
DASK-Net	0.508	20.72	0.127	5.43	0.636	26.15
MSH-Net	**0.257**	**10.65**	**0.127**	**5.43**	**0.384**	**16.08**

V. DISCUSSION

A. Comparison of Training Efficiency

The network parameters and training cost mainly come from three parts: privilege model training, hallucination model training, and ensemble fine-tuning. Among them, the parameters and cost of fine-tuning are the same for all models. Thus, we mainly compare the cost of privilege and hallucination model training. We calculated the model parameters and training FLOPs via the PyTorch-OpCounter, which is a common tool used to analyze the model parameters and operations [50], [51].

As shown in Table X, the MSH-Net has the least parameters (0.384M) and training cost (16.08M FLOPs). Specifically, during the privilege model training, the conventional methods are two-stage. In contrast, MSH-Net takes the pretrained multimodal as the privilege model directly, saving nearly half of the parameters and FLOPs. Besides, during the hallucination model training, ADD-Net, TAKD-Net, and DH-ADNet introduce extra parameters and additional costs due to the distillation methods. In contrast, MSH-Net, Hall-Net, MARS-Net, and DASK-Net only have the parameters and FlOPs of the hallucination model. This also reduces the computational overhead. Finally, it is worth noting that the parameter and FLOPs of the fine-tuning of all models are 0.96e-3 and 3.84e-3, respectively. This is negligible.

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TABLE XI
CROSS MODALITY PERFORMANCE OF MDL-CNN-NETS THAT ARE IMPLEMENTED WITH DIFFERENT FUSION STRATEGIES ON THE HUSTON2013 WITH THE HS-LIDAR COMBINATION

Method	Training Modalities	Testing Modalities	OA(%)	AA(%)	Ka
W/o LiDAR					
Decision	HS,LiDAR	HS	10.40	9.05	3.12
Cross	HS,LiDAR	HS	**47.10**	**49.20**	**46.51**
W/o HS					
Decision	HS,LiDAR	LiDAR	8.63	6.66	0.01
Cross	HS,LiDAR	LiDAR	**12.45**	**13.54**	**9.81**

B. Generalization Capacity to Unexplored Modalities

This article considers the important modalities used in multimodal remote sensing learning, including HS, MS, SAR, LiDAR, and DSM images. They can be grouped into four information sources related to the surface characteristics: spectrum (HS, MS), geometry (MS), dielectric properties (SAR), and height (LiDAR, and DSM), respectively. The proposed solution can be generalized to deal with other related modalities, such as RGB, Infrared, and PAN images. On the one hand, they can also be grouped into the four information streams. Specifically, RGB and Infrared modalities are the subset of the MS modality and contain the spectrum and geometry information [33]. PAN modality contains geometry information of surface objects [52]. On the other hand, our experiments have included both the isomorphic combinations, such as HS-MS, and heterogeneous combinations, such as HS-LiDAR, from the four information sources and verified the effectiveness of the proposed MSH-Net for them. Thus, we can expect the proposed method to be generalizable to these unexplored modalities

C. Existence of Transferable Shared Knowledge Among Highly Heterogeneous Modalities

Theoretically, consistency and complementarity are two essential characteristics of multimodality learning. The semantic consistency among multiple modalities determines that they must contain shared information no matter how different they are [53], [54]. The key problem is how to extract shared information, especially when the modalities are highly heterogeneous. This is a fundamental problem concerned by multimodal fusion [55], [56].

Although many paradigms for multimodal fusion have been proposed [5], [6], [7], [8], [9], [48], the measurement of shared information tends to be empirical. Specifically, the knowledge shared among different modalities should be robust to the absence of modalities. Thus, we evaluate the performance of the multimodal model by randomly dropping modality input and the model that extracted the shared information more effectively should perform better than those that do not. In detail, we simulate the modality missing scenarios by randomly setting one of the multimodal inputs as zeros. Then, with the feature extract network of MDL-CNNs-Net, we consider two fusion strategies, cross fusion, and decision fusion, where cross fusion is the original fusion strategy used in MDL-CNNs-Net. Decision fusion is the baseline method that only considers the modality-specific information by averaging their prediction outputs [54]. This can help evaluate the effectiveness of extracting shared knowledge. To evaluate their performance on highly heterogeneous modality data, we conduct experiments on Huston2013 with the HS-LiDAR combination.

As shown in Table XI, cross fusion outperforms decision fusion in the cross-modality tests in both "W/o LiDAR" and "W/o HS" settings. This indicates that the cross fusion model extracts shared knowledge beyond the specific knowledge of the decision fusion model in highly heterogeneous data.

VI. CONCLUSION

In this article, we propose a general framework, called MSH-Net, to assist the multimodal remote sensing image classification with missing modalities. Specifically, the MSH-Net guides the hallucination model to learn the modality-shared knowledge from the multimodal model directly. This helps preserve the cross-modal interaction among multimodal data and improves the scalability of the scenario with multiple missing modalities. Beyond the conventional FD methods, we further develop a JAD module to transfer the modality-shared knowledge from the multimodal model to the hallucination model by matching their joint probability distributions between the representations and groundtruths of the samples. This is capable of overcoming representational heterogeneity caused by the input and structure discrepancy between the hallucinations and multimodal models while preserving the decision boundaries refined by multimodal cues. Finally, extensive experimental results on four common modality combinations verify the effectiveness of our methods.

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